

WORKSHOP 2



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Databases 2

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Project Context

This project proposes the conceptual design and basic implementation of a digital platform called Bookmate, which allows users to search, read, listen to, rate, and share books and audiobooks. The system also offers community features such as profiles, favorites lists, user tracking, and personalized recommendations, as well as a monetization model through subscription plans.

The system architecture is based on the combined use of SQL (PostgreSQL) and NoSQL (MongoDB) databases, allowing for efficient management of structured (users, subscriptions) and semi-structured (reviews, books) data.

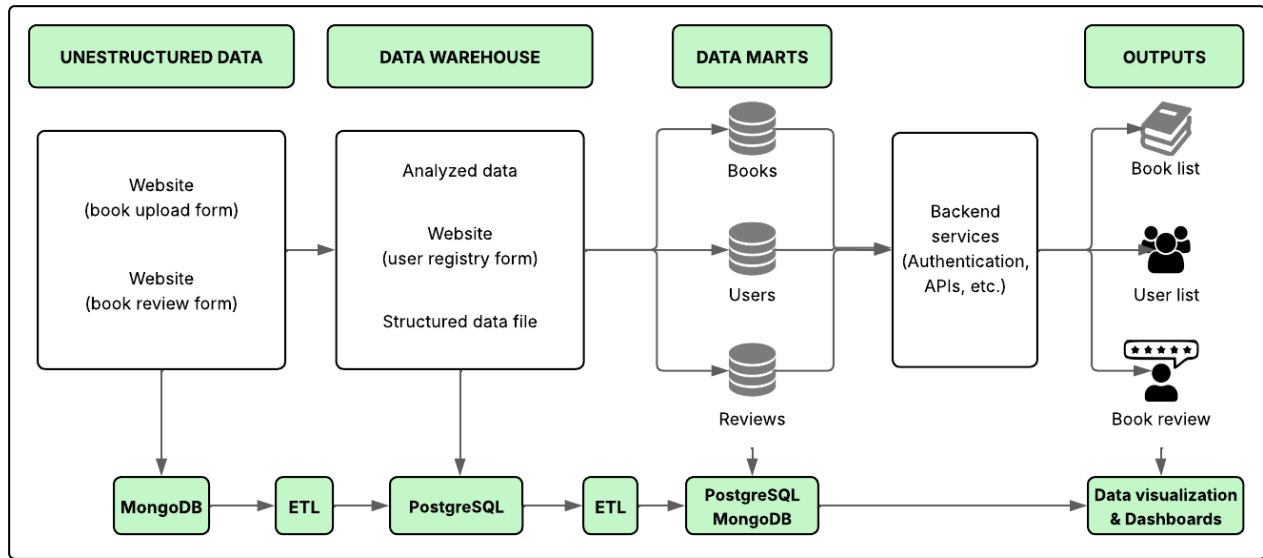
- **Project Scope**

The scope of this project focuses on the design of the data architecture, the definition of functional requirements, and the implementation of a minimum set of key functionalities. These include:

- Book/audiobook uploading and searching
- User profile creation
- Review and favorites system
- Basic Premium subscription model
- Proposed representative SQL and NoSQL queries

The project does not contemplate the full development of the frontend or advanced business logic, but rather focuses on demonstrating how a functionally coherent solution can be designed and implemented at the database level, based on a real platform.

Database Architecture



In the following link you can see the Architecture diagram: [Architecture](#)

1. Initial Data Capture (Unstructured Data Sources)

Website (book upload form) / Website (book review form): These are the user interfaces where data is initially entered. The "book upload" form allows users to upload information about books, which can include details such as title, author, cover, description, etc. The "book review" form allows users to write and submit reviews of books. This data is typically unstructured or semi-structured, so MongoDB will be used because of its flexible with schemas, which is ideal for data that can vary in structure, such as book reviews or initial book information that has not yet been fully processed or validated.

2. Structured Processing and Storage (Data Warehouse)

The ETL process converts unstructured or semi-structured data into a structured format suitable for analysis. The **Data Warehouse**, built using **PostgreSQL**, stores this data due to its strong relational capabilities and support for complex SQL queries.

Key structured data sources include:

- **Analyzed data:** Already processed information ready for advanced querying.

- **Website forms:** Structured input like user registrations (name, email, password).
- **Structured files:** Data imported from CSVs, spreadsheets, or other external structured formats.

3. Data Segmentation (Data Marts)

A second ETL process extracts data from the main PostgreSQL warehouse, transforms it for specific purposes, and loads it into data marts—smaller, focused databases optimized for quick access and specific analyses.

- **Books (MongoDB):** Book data is highly variable (e.g., multiple authors, editions, translators). MongoDB handles this diversity well through flexible JSON documents, avoiding the rigid schema and NULLs typical in relational databases.
- **Users (PostgreSQL):** User data (ID, email, roles, etc.) is structured and relational. PostgreSQL is ideal for this due to its strong ACID guarantees, support for complex relationships, and robust security for sensitive information.
- **Reviews (MongoDB):** Reviews vary in length and structure (text, ratings, tags). MongoDB efficiently stores this unstructured or semi-structured feedback at scale, making it ideal for large volumes of user-generated content.

Information Requirements

Information Type	Description
Book and Audiobook Search	Retrieve books based on filters such as title, author, genre, or popularity.
User Profile Data	Fetch user details including name, profile picture, subscription status, and preferences.
Favorites and Reading Lists	Load a user's saved books, favorite titles, and personalized reading lists.
User Reviews and Ratings	Retrieve reviews and star ratings submitted by users for a specific book or audiobook.
Followed User Activity	Show public lists, favorites, or reviews from users the person follows.
Recommendations	Retrieve book or audiobook suggestions based on reading history, social activity, and preferences.
Subscription status	Check the user's current subscription plan and access level (e.g., Premium).
Administrative Content List	Fetch books uploaded by administrators or creators, including metadata for review or editing.
Audit and Moderation Logs	Access logs of content flagged, edited, or removed for moderation purposes.
Analytics and Reports	(Future) Generate statistics on book popularity, user engagement, and feature usage.

Query Proposals

1. **Requirement FR-1.1 (SQL):** The system must allow book and audiobook searching by title, author, or genre using a search engine with combinable filters.

```
SELECT b.title, a.name AS author, g.description AS genre
FROM book b
JOIN author a ON b.author = a.author_id
JOIN genre_book gb ON b.book_id = gb.book_id
JOIN genre_type g ON gb.genre_type_id = g.genre_type_id
WHERE b.title ILIKE '%magic%'
      OR a.name ILIKE '%rowling%'
      OR g.description ILIKE '%fantasy%';
```

The purpose of this query is to allow users to search by keywords across multiple fields with flexible filters. This would help users to quickly find books or audiobooks of interest, improving discoverability and user experience.

2. **Requirement FR-2.2 (NoSQL):** The system should allow users to write reviews associated with each title and allow editing and deletion if necessary.

```
book_platform.review.insertOne({
  UUID(),
  user_id: "New Horizons",
  book_id: "0304fef4-5e2e-4e1f-9563-f3274dc1fd9b",
  rating: 5,
  comment: " This book was amazing – highly recommended!",
  createdAt: new Date(),
});
```

The purpose of this query is to allow a user to submit a review and rating for a specific book, and it contributes to the collective evaluation of content and improves recommendation quality.

3. **Requirement FR-3.1 (SQL):** The system must allow the creation and editing of a user profile with a photo, display name, and short description.

```
INSERT INTO user (user_id, password, name, email, profile_picture, is_premium,
user_type, created_at)
VALUES (
  gen_random_uuid(),
  '$2b$10$7nDfF3uH1qAn2AymzDqXz.LwPS1YVLz6obGZuhmT.yzGU6/10hzHq',
  'Jane Doe',
  'jane@example.com',
  'https://cdn.app/pics/jane.jpg',
  false,
  (SELECT user_type_id FROM user_type WHERE description = 'Reader'),
  CURRENT_TIMESTAMP
);
```

The purpose of this query is to register a new user and create their personalized profile. It enables users to have an account and other user-specific features like reading lists, follows, and recommendations.

4. **Requirement FR-4.1 (NoSQL):** The system must allow profiles with the Administrator or Creator role to upload book and audiobook files with their metadata.

```
book_platform.book.insertOne({
  UUID(),
  title: "New Horizons",
  author: "9b57fe43-a03f-4a15-8ce5-8e5f7b01b341",
  genres: "57c1488f-d781-4eac-9d6f-7e750675fabf ",
  synopsis: "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis vestibulum nibh quis turpis feugiat dictum. Etiam eget faucibus sem. Sed non fringilla odio, in sollicitudin risus. Maecenas in neque ut odio iaculis vestibulum at eu tellus. Etiam tempus tincidunt dolor sed euismod. Integer dapibus fringilla ultricies. Sed risus sapien, semper viverra urna vitae, consectetur varius nunc.",
  coverImage: "https://cdn.app/pics/jane.jpg",
  language: "English",
  IsAudiobook: true,
  fileUrl: https://cdn.app/pics/jane.jpg,
  createdAt: new Date(),
  updatedAt: new Date(),
  uploadedBy: "bad0a5df-257b-409c-95a4-f7cc52e5a65d"
});
```

The query adds a new book to the catalog with basic metadata, linked to the creator's user ID. This expands available content and supports the creator-admin workflow for content management.

5. **Requirement FR-6.1 (SQL):** The system must allow users to purchase a Premium plan with payment management and access control to exclusive content.

```
INSERT INTO subscription (subscription_id, user_id, plan_type, start_date,
end_date, payment_status)
VALUES (
  gen_random_uuid(),
  'bad0a5df-257b-409c-95a4-f7cc52e5a65d',
  (SELECT plan_type_id FROM plan_type WHERE name = 'Premium'),
  CURRENT_DATE,
  CURRENT_DATE + INTERVAL '30 days',
  (SELECT payment_status_type_id FROM payment_status_type WHERE description =
  'Paid')
);
```

The query registers a new subscription for a user, specifying the plan and payment status. It also controls access to paid content and helps analyze revenue and user upgrade trends.

```
SELECT plan_type, start_date, end_date, payment_status
FROM subscription s JOIN plan_type p ON s.plan_type = p.plan_type_id
JOIN payment_status_type pst ON s.payment_status = pst.payment_status_type_id
WHERE s.user_id = 'current-user-id'
      AND CURRENT_DATE BETWEEN s.start_date AND s.end_date;
```

This query verifies if a user currently has an active Premium subscription and controls access to exclusive content and features based on the user's subscription status.

- **SQL + NoSQL Strategy: Complementary Use**

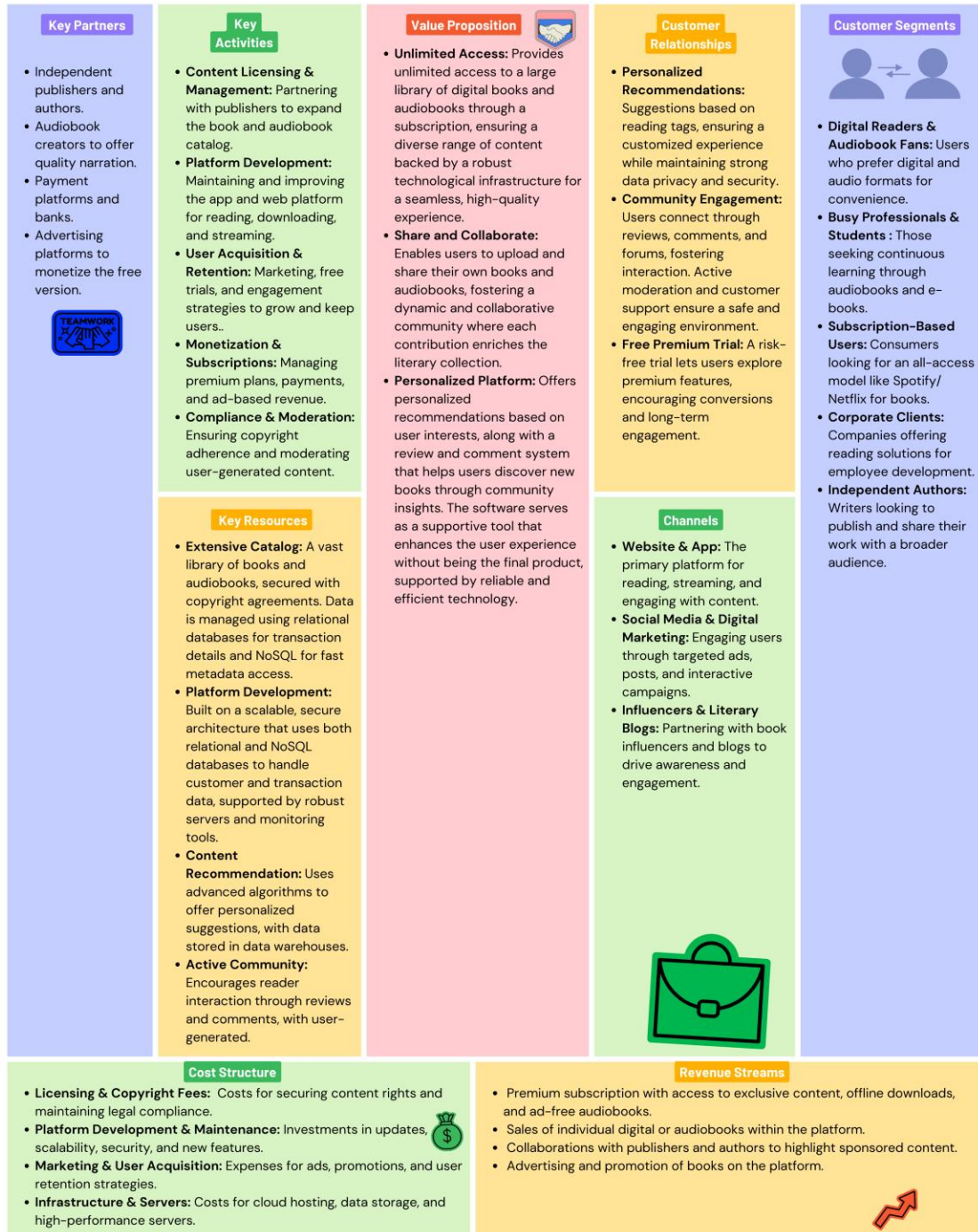
PostgreSQL is used to manage structured, relational data that requires consistency and integrity. Its support for SQL, constraints, and strong ACID properties makes it ideal for securely handling user information, subscriptions, and other well-defined entities.

MongoDB is chosen for storing flexible and variable data structures, such as books and reviews. As a document-oriented NoSQL database, it handles nested and inconsistent data efficiently, allowing rapid development and scalability without enforcing a rigid schema.

Business Model

Reading platform based on Bookmate

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In the following link you can see the Canva Model diagram: [Business model - Bookmate](#)

System Requirements

- **Functional Requirement**

1. **Audiobooks and reading module:**

Code	Requirement
FR-1.1	The system must allow book and audiobook searching by title, author, or genre using a search engine with combinable filters.
FR-1.2	The system must offer an online reading viewer that loads the text without prior download and quick page turning.
FR-1.3	The system must enable pause, fast-forward, and rewind controls for streaming audiobook playback, ensuring seamless continuity.
FR-1.4	The system must allow downloading complete books and audiobooks for offline reading and listening, storing them locally with space management.
FR-1.5	The reading viewer should allow users to adjust the font size and background theme (light/dark, sepia, etc.) to improve the reading experience.

2. **Reviews, favorites and comments:**

Code	Requirement
FR-2.1	The system should allow users to rate books and audiobooks using a star system (1-5).
FR-2.2	The system should allow users to write reviews associated with each title and allow editing and deletion if necessary.
FR-2.3	The system must display the list of existing reviews for the user to see and decide whether to read or listen to a title.
FR-2.4	The system must allow titles to be marked as favorites and display a personal list of favorites accessible from the profile.

3. **Community and profiles:**

Code	Requirement
FR-3.1	The system must allow the creation and editing of a user profile with a photo, display name, and short description.
FR-3.2	The system must allow users to follow other users and view their public activity (reviews and lists).
FR-3.3	The system should display the favorites of followed users to facilitate discovery.
FR-3.4	The system must generate notifications when one user is followed by another.
FR-3.5	The system must maintain a list of followers and followed for each user and show follow/unfollow actions.

4. **Books management and uploading:**

Code	Requirement
FR-4.1	The system must allow profiles with the Administrator or Creator role to upload book and audiobook files with their metadata.

FR-4.2	The system must allow Administrators or Creators to edit metadata (cover, synopsis, author, tags) of any title they have uploaded.
FR-4.3	The system shall allow an Administrator to remove content that violates rules and record the action in an audit log.

5. Recommendations and discoveries:

Code	Requirement
FR-5.1	The system should generate personalized recommendations based on the user's reading history and ratings.
FR-5.2	The system should suggest books read by friends (followed users) that the user has not yet read.
FR-5.3	The system should offer curated thematic collections (by the community or administrators) and navigable from a “Discover” section.

6. Monetization and subscriptions:

Code	Requirement
FR-6.1	The system must allow users to purchase a Premium plan with payment management and access control to exclusive content.
FR-6.2	For Premium users, the system must enable unrestricted downloading of books and audiobooks.
FR-6.3	For Premium users, the system must play audiobooks without ads or promotional interruptions.

• Non-Functional Requirements

1. Performance

Code	Requirement
FNR-1.1	The system should allow book searches with response times of less than 2 seconds for simple queries.
FNR-1.2	Loading profiles and favorites lists should complete in less than 1 second under normal conditions.

2. Scalability

Code	Requirement
FNR-2.1	The architecture should allow MongoDB to scale horizontally to handle large volumes of reviews and books in the future.
FNR-2.2	The solution should be able to handle at least 10,000 active users without significant performance degradation.

3. Availability

Code	Requirement
FNR-3.1	The system should be available at least 95% of the time during development and testing.

4. Security

Code	Requirement
FNR-4.1	User passwords should be securely stored using hashing techniques (e.g., bcrypt).
FNR-4.2	Access to administrative functions should be restricted by roles and authentication.

5. Maintainability

Code	Requirement
FNR-5.1	SQL and NoSQL queries should be documented to facilitate understanding by new developers.

6. Compatibility

Code	Requirement
FNR-6.1	The system must be compatible with modern browsers (Chrome, Firefox, Edge).
FNR-6.2	The database must be easily replicable in Linux and Windows environments for testing.

7. Portability

Code	Requirement
FNR-7.1	The system must be able to be installed locally or on cloud services (such as Heroku or Railway) without requiring complex configurations.
FNR-1.2	Loading profiles and favorites lists should complete in less than 1 second under normal conditions.

User Stories

1. Reviews, favorites and comments

Title: Book search	Priority: High
User Story: As a user, I want to search for books and audiobooks by their title, author or genre to find content of interest.	
Acceptance Criteria: - A search bar is available on the main page or catalog page for entering queries. - When a user types a book or audiobook title, relevant results matching the title are displayed. - When a user enters an author's name, the system returns all books or audiobooks associated with that author. - When a user selects or types a genre (e.g., "Science Fiction", "Biography"), matching content is displayed. - The user can combine title, author, and genre in the same search to narrow down results.	

- The search works regardless of capitalization (e.g., "Harry Potter" = "harry potter").
- If no results are found, the system displays a message like “No results found” and suggests possible corrections.
- Search results are displayed within 2 seconds after submitting a query, assuming normal internet conditions.

Title: Online reading	Priority: High
User Story: As a user, I want to read books online without needing to download them.	
Acceptance Criteria: • The system provides an online reading viewer accessible directly from the book’s detail page. • The user can start reading the book without having to download any file to their device. • The first page of the book loads after opening the viewer. • The user can navigate through the book pages using next/previous controls or scrolling, without delays. • If the book cannot be loaded, a meaningful error message is shown to the user, and the system does not crash.	

Title: Audiobook streaming	Priority: High
User Story: As a user, I want to stream audiobooks without interruptions.	
Acceptance Criteria: • The user can play an audiobook directly without needing to download it first. • The user can pause, play, rewind, and fast-forward the audiobook seamlessly during streaming. • The system adjusts playback quality based on network speed to avoid interruptions. • If the user pauses or leaves the player, the system remembers the last listened position and resumes from there. • If there is a streaming failure (e.g., connection loss), a clear error message is shown and playback resumes automatically once the connection is restored. • Premium users stream audiobooks without ads or promotional interruptions.	

Title: Book download	Priority: Medium
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User Story: As a user, I want to save books or audiobooks to read to my user profile to read online.
Acceptance Criteria: • Each book and audiobook includes a clearly visible “Save” button or option on its detail page. • Users can choose to save either the book text or the audiobook version, depending on availability. • The system displays saved books and audiobooks and notifies the user when a new item is saved. • Users can view and manage saved content, including the ability to delete items to free up space. • Saved content remains accessible even after the app is closed and reopened, as long as the user is logged in.

Title: Reading experience	Priority: Medium
User Story: As a user, I want to adjust the font size and background to enhance my reading experience.	
Acceptance Criteria: • The reading interface provides a control (e.g., slider, buttons) to increase or decrease the font size. • The user can switch between different font sizes and different background themes such as light or dark. • Changes in font size or background theme are applied immediately in the reading view without reloading the page and are stored as preferences. • Users can reset to default font size and background with a single action (e.g., "Reset to default" button).	

2. Community and profiles

Title: Rating creation	Priority: High
User Story: As a user, I want to be able to rate books and audiobooks with stars to share my opinion with the community.	
Acceptance Criteria: • A star-based rating interface (e.g., 1 to 5 stars) is visible on each book and audiobook’s detail page.	

- Each user can submit only one rating per book or audiobook, which can be updated later.
- The submitted rating is stored and persists across sessions (e.g., it remains visible when the user returns).
- Only authenticated users can rate content; unauthenticated users are prompted to log in when attempting to rate.

Title: Review reading

Priority: High

User Story: As a user, I want to be able to read reviews from other readers before deciding if I want to read a book or not.

Acceptance Criteria:

- Each book or audiobook detail page includes a visible section where user reviews are displayed.
- Reviews can be read by all users, including unauthenticated visitors.
- Each review displays the reviewer's username (or display name), submission date, star rating (if given), and the review text.
- Users can sort reviews by most recent, highest rated, or most helpful.
- The total number of reviews for each title is displayed next to the reviews section or rating summary.

Title: Favorite list

Priority: High

User Story: As a user, I want to be able to save books and audiobooks on my favorites list so I can access them easily.

Acceptance Criteria:

- Each book and audiobook detail page includes a clearly visible button (e.g., a heart or bookmark icon) to mark the item as a favorite.
- Clicking the favorite button adds the book or audiobook to the user's personal favorites list.
- Users can view their complete favorites list from their profile or a dedicated "Favorites" section.
- Only logged-in users can add or view favorites; unauthenticated users are prompted to log in.
- The favorites list can optionally be sorted by title, date added, or content type (book vs. audiobook).

3. Books management and uploading

Title: Profile creation	Priority: High
User Story: As a user, I want to be able to create a personal profile with my picture, name and a short description to personalize my account.	
Acceptance Criteria: • Upon account creation or first login, the system provides access to a profile setup or editing page. • The user can upload a profile picture from their device (common formats like JPG, PNG supported). • The user can enter a display name that will be shown on public-facing areas (e.g., reviews, profile page). • Users can update their picture, name, and description at any time from the profile settings. • Users can preview changes before saving, and a confirmation message is shown after successful updates. • The user's profile (picture, name, and description) is visible to other users in social or community areas (e.g., review sections, follower lists).	

Title: Following	Priority: Medium
User Story: As a user, I want to be able to follow other readers to see their recommendations and reading lists.	
Acceptance Criteria: • Each user profile includes a visible "Follow" button for authenticated users to follow others. • Users can unfollow others at any time by clicking the same button, which toggles to "Unfollow" after following. • The user being followed receives a notification when someone follows them. • Only public profiles or public activity is visible to followers; private content is not shown unless shared intentionally. • Follow/unfollow functionality works seamlessly on mobile, tablet, and desktop devices.	

Title: Favorite list of followers	Priority: Low
User Story: As a user, I want to be able to see the favorite books and audiobooks of the users I follow to discover new titles.	

Acceptance Criteria:

- Each followed user's public profile includes a section displaying their favorite books and audiobooks.
- Only favorites marked as public by the followed user are shown; private favorites are hidden.
- If a followed user adds or removes a favorite, the change is reflected in real time or after a page refresh.
- Only logged-in users can follow others and view their favorites.

Title: Following notification**Priority: Medium****User Story:** As a user, I want to receive notifications when someone follows me.**Acceptance Criteria:**

- When a user is followed by another user, the system generates a notification for the followed user.
- Notifications are accessible from a dedicated notifications area in the user's profile or header menu.
- The notification remains available in the user's notification list until marked as read or deleted.

Title: Follow users**Priority: Medium****User Story:** As a user, I want to be able to follow and be followed by other users to connect with other readers.**Acceptance Criteria:**

- Each user profile includes a visible button that allows authenticated users to follow or unfollow others.
- The number of followers and the number of users followed are displayed on the user's profile.
- Users receive a notification when someone follows them.
- The follow button updates appropriately depending on the user's current relationship with another user (e.g., shows "Follow" or "Unfollow").

4. Recommendations and discoveries**Title: Book uploading****Priority: High**

User Story: As an administrator or creator, I want to be able to upload books and audiobooks to share with the community.

Acceptance Criteria:

- Only users with the "Administrator" or "Creator" role can access the upload interface.
- The system provides a dedicated form for uploading book and audiobook files, accessible from the dashboard or profile.
- The upload form requires metadata fields such as title, author, cover image, synopsis, tags, and genre.
- The system accepts valid file formats for books (e.g., PDF, EPUB) and audiobooks (e.g., MP3, M4A).
- The system validates all required fields and files before submission and shows appropriate error messages if validation fails.
- Uploaded books and audiobooks become visible to all users once approved or published, depending on platform policies.

Title: Book details editing

Priority: High

User Story: As an administrator or creator, I want to be able to edit the details of the books (cover, synopsis, author and tags).

Acceptance Criteria:

- Only users with the "Administrator" or "Creator" role can access the editing interface for books they have uploaded (or all books, for admins).
- The editing form includes fields for cover image, synopsis, author name, and tags.
- Updated book information is reflected in the public book detail page immediately (or after admin approval, depending on workflow).
- If saving fails (e.g., due to connectivity or validation issues), the system displays an appropriate error message and does not apply partial changes.

Title: Content deletion

Priority: High

User Story: As an administrator, I want to be able to delete content that infringes the norms to assure the quality of the platform.

Acceptance Criteria:

- Only users with the "Administrator" role can access the delete option for books and audiobooks.

- Each book or audiobook includes a visible “Delete” or “Remove” button or action in the admin interface.
- A confirmation dialog appears before deletion, warning the administrator that the action is irreversible.
- Once deleted, the content is no longer visible to users in search results, lists, or profiles.
- The admin dashboard allows searching, filtering, or sorting content to locate potential violations easily.

5. Monetization and subscriptions

Title: Recommendations based on own readings	Priority: Medium
User Story: As a user, I want to receive recommendations of books and audiobooks based on my readings and reviews.	
Acceptance Criteria: • A "Recommended for you" section is available on the user’s homepage or dashboard after logging in. • The system generates recommendations based on books and audiobooks the user has read or listened to. • Recommended items are of similar genres, authors, or themes to those the user has previously engaged with. • Clicking a recommendation takes the user to the book or audiobook’s full detail page. • Recommendations are only shown to logged-in users, based on their individual activity.	

Title: Recommendations based on friends readings	Priority: Medium
User Story: As a user, I want the platform to suggest new books based on what my friends are reading.	
Acceptance Criteria: • A section labeled “Books Your Friends Are Reading” (or similar) is available in the user’s dashboard or discovery page. • The system suggests books and audiobooks that have been read, favorited, rated, or reviewed by users the current user follows. • The list updates dynamically as followed users read or interact with new content. • Only logged-in users can view friend-based recommendations and follow others.	

Title: Exclusive content for premium users	Priority: High
User Story: As a premium user, I want to be able to access a premium plan to unlock exclusive content and advanced functionalities.	
Acceptance Criteria: • Users can view a clear description of available premium plans, including pricing, benefits, and duration. • Users can subscribe to a premium plan using secure payment methods (e.g., credit card, PayPal). • Upon successful payment, the user account is upgraded to “Premium” and gains immediate access to premium features. • Premium users can access content marked as "Premium" (e.g., exclusive books, audiobooks, or early releases). • If payment fails or there is a technical issue during subscription, the system displays a clear error message and provides a support contact.	

Title: Downloads for premium users	Priority: High
User Story: As a premium user, I want to be able to download books without restrictions to read them offline.	
Acceptance Criteria: • Premium users can download any number of available books without hitting a download limit or quota. • A clearly labeled “Download” button appears on all book detail pages when the user has a premium subscription. • The download starts promptly after clicking, with a progress indicator and a message confirming completion. • Downloaded files are stored in a protected format or location that prevents unauthorized distribution or modification. • Downloaded books are accessible for offline reading through the app or platform interface. • Downloaded books remain accessible as long as the user’s premium subscription is active and the book is not removed from the platform.	

Title: Advertising for premium users	Priority: High
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User Story: As a premium user, I want to be able to enjoy audiobooks without advertising for a better experience.

Acceptance Criteria:

- Audiobooks play without any ads or promotional interruptions for users with an active premium subscription.
- Once a user subscribes to the premium plan, ads are automatically removed from audiobook playback—no manual settings required.
- Playback transitions are seamless, without buffering or delays where ads would normally appear.
- In the rare case of a playback or role check error, a fallback message is displayed, and the user is not shown ads if premium status is valid.

Entity Relationship Model

The main components of the system were identified based on user stories and platform functionalities. These include users, books/audiobooks, reviews, subscriptions, reading lists, favorites, notifications, and user interactions (e.g., follows, progress tracking).

Key entities were extracted from the components. The entities include: *User*, *Book*, *Review*, *Subscription*, *Reading_List*, *Reading_List_Item*, *Favorite*, *Notification*, *Follow*, *User_Book* (progress tracking) and lookup tables such as *User_Type*, *Plan_Type*, *Genre_Type*, *Payment_Status_Type*, *Notification_Type*. A dedicated *Audiobook* entity was introduced to represent the audio version of a book as an optional one-to-one extension, allowing for specific audio attributes like duration and narrator to be stored independently while maintaining referential integrity with the *Book* entity.

Each entity was assigned specific attributes, such as title, author, genre, language, cover_image for *Book*. Also, *User* includes name, profile_picture, is_premium, and user_type_id. Data types like Timestamps were used for attributes such as created_at and updated_at.

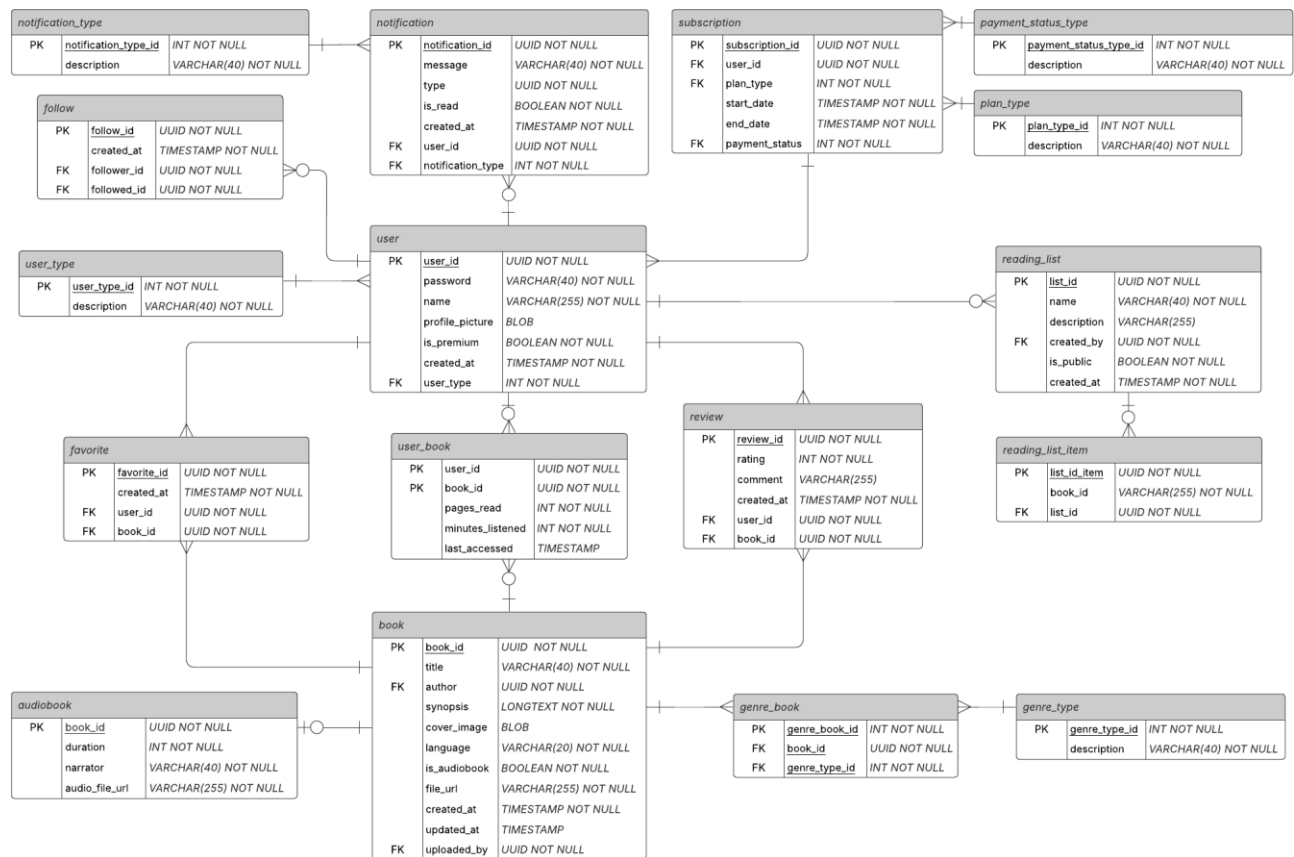
book		
PK	book_id	UUID
	title	VARCHAR(40)
FK	author	UUID
FK	genre	UUID
	synopsis	LONGTEXT
	cover_image	BLOB
	language	VARCHAR(20)
	is_audiobook	BOOLEAN
	file_url	VARCHAR(255)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP
FK	uploaded_by	UUID

user		
PK	user_id	UUID
	password	VARCHAR(40)
	name	VARCHAR(255)
	profile_picture	BLOB
	is_premium	BOOLEAN
	created_at	TIMESTAMP
FK	user_type	INT

Relationships between entities were then established to reflect real-world interactions within the platform. These included one-to-many relationships, such as a user creating multiple reviews, and many-to-many relationships, such as books belonging to multiple genres or users tracking progress across different books. All many-to-many relationships were later normalized into separate junction tables like *Genre_Book*, *User_Book*, and *Reading_List_Item*.

user_book		
PK	user_book	UUID
FK	user_id	UUID
	pages_read	INT
FK	book_id	UUID

An initial ER model was drafted to visualize the core entities and their interactions. This version was refined by splitting many-to-many relations and formalizing optional one-to-one structures, such as the *Audiobook* entity, which is modeled as 1:1 extensions of book, using the same book_id as primary and foreign key, which better reflects their dependency and improves referential clarity. Data types were explicitly defined using UUIDs for entity identifiers, along with strong typing for timestamps, booleans, and text fields.



In the following link you can see the Entity Relationship Model: [E-R Model](#)